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Inventor(s)	Beebe; David J. et al.

Method and device for creating organotypic lumen models in a microwell plate

Abstract

A method and device of creating a lumen model in a microwell plate defining a well is provided. A rod is inserted through the well of the microwell plate and the well is filled with a polymerizable material. The material is polymerized in the well. The rod is removed from the well of the microwell plate such that the polymerized material defines a lumen.

Inventors: **Beebe; David J. (Monona, WI), McMinn; Patrick H. (Madison, WI)**

Applicant: **Wisconsin Alumni Research Foundation** (Madison, WI)

Family ID: **80356151**

Assignee: **Wisconsin Alumni Research Foundation (Madison, WI)**

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Primary Examiner: Levkovich; Natalia

Attorney, Agent or Firm: Boyle Fredrickson, S.C.

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a division of U.S. application Ser. No. 17/004,268, filed Aug. 27, 2020, the entirety of which is incorporated herein by reference.

FIELD OF THE INVENTION
(1) This invention relates generally to microfluidics, and in particular, to a method of creating organotypic lumen models in a microwell plate.
BACKGROUND AND SUMMARY OF THE INVENTION
(2) Advances in modern healthcare in the last few decades have, in part, led to better quality of life and higher life expectancy worldwide. However, mortality due to communicable and non-communicable diseases, such as drug-resistant tuberculosis, cardiovascular diseases, and cancers, continues to rise. A concerted effort from multiple fronts is needed to lessen global disease burden, including better preventive care (e.g. lifestyle education), improved access to healthcare services, affordable medical diagnosis and treatment, and continued growth of biomedical research. In the context of biomedical research, developing relevant tissue and organ models to better understand the pathophysiology of human diseases is of great importance to therapeutic discovery and implementation.
(3) Two-dimensional (2D) in vitro culture models have formed the cornerstone of biomedical research for more than a century. Despite their ease-of-use and high experimental tractability, these models fail to recapitulate the three-dimensional (3D) tissue- and organ-level structure-function relationships observed in vivo. These limitations have led to the development of 3D culture techniques, including cells embedded in or on top of natural or synthetic hydrogels. Continued innovation has also led to organotypic culture, wherein self-assembled cellular clusters or organoids better mimic organ-level physiology. Compared with 2D cell culture, 3D techniques enable deeper analysis of the molecular mechanisms governing the pathophysiology of different diseases and can be valuable preclinical tools for drug testing. Nonetheless, current 3D models remain limited in their ability to capture crosstalk between multiple cell types and mechanical cues in the microenvironment that contribute to cellular mechanotransduction (e.g. fluid shear stress), which are important factors in disease progression. Mouse models could be suitable alternatives as they provide physiologically relevant microenvironments. However, mouse models have low experimental tractability and the evolution of the xenografted disease could be mouse-specific rather than patient-specific5, questioning the validity of such models.
(4) Microfluidic cell culture, i.e. the synergy of tissue engineering with microscale physics, has enabled the development of microphysiological systems or organs-on-chips. These systems enable multi-culture of different cell types at physiological length scales, generation of stable biochemical

gradients, and controlled fluid transport, among other unique capabilities. Microphysiological systems have been used to mimic many human organ systems, including the heart, brain, kidney, liver, lung, pancreas, and circulation. Importantly, organs-on-chips have demonstrated utility for unraveling basic mechanisms of disease and for predicting patient response to drugs in the context of translational medicine.

(5) Lumens are tubular structures that are pervasive throughout all body systems, forming blood vessels that regulate nutrient and waste exchange in tissues to mammary ducts for the production of milk to intestinal tracts for nutrient absorption. Lumens are, perhaps, the earliest 3D geometry to develop during the process of embryonic neural tube formation. In major diseases such as cancer, ductal dysfunction in the breast and prostate can initiate tumorigenesis, and in later stages, the conditioning of normal blood and lymphatic vessels to a tumor-associated phenotype enables the spread of cancer cells to secondary sites in the body. Similarly, the stiffening and narrowing of blood vessels in atherosclerotic vascular disease can lead to complications in major organs including the heart, brain and kidney. Therefore, the importance of capturing luminal geometry in disease models goes beyond aesthetics to mimicking in vivo structure-function relationships.

(6) Therefore, it is a primary object and feature of the present invention to provide a method and device for creating lumen models in a microwell plate.

(7) It is a further object and feature of the present invention to provide a method and device for creating lumen models in a microwell plate that may be automated so as to allow for high-throughput fabrication.

(8) It is a still further object and feature of the present invention to provide a method and device for creating lumen models in a microwell plate that is simple and inexpensive to implement.

(9) In accordance with the present invention, a method of creating a lumen model in a microwell plate defining a well is provided. The method includes the steps of inserting a rod through the well of the microwell plate and filling the well with a polymerizable material. The material in the well is polymerized and the rod is removed from the well of the microwell plate such that the polymerized material defines a lumen.

(10) The rod may have a circular cross-section and may be magnetic or be fabricated from a polymer. The rod may be a first rod and the lumen may be a first lumen. As such, a second rod may be inserted through the well of the microwell plate and removed from the well of the microwell plate after the material in the well is polymerized such that the polymerized material defines a second lumen. The first and second lumens may be generally parallel or perpendicular. In addition, the first and second lumens may extend along corresponding axes that are axially spaced or the first lumen may communicate with the second lumen.

(11) In accordance with a further aspect of the present invention, a method of creating a lumen model in a microwell plate defining a plurality of wells. Each of the plurality of wells is defined by at least one sidewall. The method includes the step of inserting at least one rod through first and second openings in each of the at least one sidewall defining the plurality of wells and through the plurality of wells. At least a portion of the plurality of wells are filled with a polymerizable material and the material is polymerized in the at least a portion of the plurality of wells. The at least one rod is removed from the first and second openings in each of the at least one sidewall defining the plurality of wells and from the plurality of wells such that the polymerized material in the at least a portion of the plurality of wells define first lumens in the at least a portion of the plurality of wells.

(12) Third and fourth openings may be provided in each of the at least one sidewall defining the plurality of wells. The at least one rod may be inserted through third and fourth openings in each of the at least one sidewall defining the plurality of wells and through the plurality of wells. The at least one rod is removed from the third and fourth openings in each of the at least one sidewall defining the plurality of wells and from the plurality of wells such that the polymerized material in the at least a portion of the plurality of wells define second lumens in the plurality of wells. The first and second lumens in each well of the at least a portion of the plurality of wells may be

generally parallel or generally perpendicular. In addition, the first and second lumens in each well of the at least a portion of the plurality of wells may extend along corresponding axes which are axially spaced. Further, the first lumen in each well of the at least a portion of the plurality of wells may communicate with the second lumen in each well of the at least a portion of the plurality of wells.

(13) It is contemplated for the steps of inserting the at least one rod through the first and second openings in each of the at least one sidewall defining each well of the plurality of wells and through the plurality of wells and inserting the at least one rod through the third and fourth openings in each of the at least one sidewall defining each well of the plurality of wells and through the plurality of wells to occur simultaneously.

(14) In accordance with a still further aspect of the present invention, a device is provided for creating a lumen in a microenvironment. The device includes a plurality of rods and a microwell plate defining a plurality of wells arranged in rows and columns. Each of the plurality of wells defined by at least one sidewall having first and second openings therein. Each rod is selectively moveable between a first position wherein each rod is removed from corresponding first and second openings in each of the at least one sidewall defining the plurality of wells and a second position wherein each rod is inserted through the corresponding first and second openings in each of the at least one sidewall defining the plurality of wells. A fixture is interconnected to the plurality of rods. The fixture is configured for moving the plurality of rods between the first and second positions.

(15) Each of the plurality of rods may be parallel to each other. In addition, the at least one sidewall defining the plurality of wells has third and fourth openings therein. The plurality of rods may be a first plurality of rods and the device may include a second plurality of rods. Each rod of the second plurality of rods is selectively moveable between a first position wherein each rod of the second plurality of rods is removed from the third and fourth openings in each of the at least one sidewall defining the plurality of wells and a second position wherein each rod of the second plurality of rods is inserted through the third and fourth openings in each of the at least one sidewall defining the plurality of wells.

(16) The fixture may be interconnected to the second plurality of rods. The fixture may be configured for moving the second plurality of rods between the first and second positions. Alternatively, the fixture may be a first fixture and the device may include a second fixture interconnected to the second plurality of rods. The second fixture is configured for moving the second plurality of rods between the first and second positions.

(17) The first plurality of rods may lie in a first plane and the second plurality of rods may lie in a second plane spaced from the first plane. The first plurality of rods may be parallel or transverse to the second plurality of rods. Each of the first plurality of rods travels along a path between the first and second positions of the first plurality of rods and each of the second plurality of rods travels along a path between the first and second positions of the second plurality of rods. The path of each of the first plurality of rods is interested by the paths of the second plurality of rods.

(18) Each of the plurality of rods may have a circular cross-section. In addition, each of the plurality of rods may magnetic or may be fabricated from a polymer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings furnished herewith illustrate a preferred construction of the present invention in which the above advantages and features are clearly disclosed as well as others which will be readily understood from the following description of the illustrated embodiment.

(2) In the drawings:

(3) FIG. **1** is schematic view of a system for effectuating a methodology of fabricating lumen models in a microwell plate in accordance with the present invention;
(4) FIG. **2** is a cross-sectional view of the system of the present invention taken along line **2-2** of FIG. **1**;
(5) FIG. **3** is a top plan view of the system of FIG. **3**;
(6) FIG. **4** is a cross-sectional view of the system of the present invention, similar to FIG. **2**, showing a lumen model fabricated in the microwell plate;
(7) FIG. **5** is a top plan view of the system of the present invention, similar to FIG. **3**, showing a lumen model fabricated in the microwell plate;
(8) FIG. **6** is schematic view of an alternate configuration of a system for effectuating a methodology of fabricating lumen models in a microwell plate in accordance with the present invention;
(9) FIG. **7** is a cross-sectional view of the system of the present invention taken along line **7-7** of FIG. **6**;
(10) FIG. **8** is a top plan view of the system of FIG. **7**;
(11) FIG. **9** is a cross-sectional view of the system of the present invention, similar to FIG. **7**, showing first and second lumen models fabricated in the microwell plate;
(12) FIG. **10** is a top plan view of the system of the present invention, similar to FIG. **8**, showing the first and second lumen models fabricated in the microwell plate;
(13) FIG. **11** is a cross-sectional view of the system of the present invention taken along line **11-11** of FIG. **10**;
(14) FIG. **12** is schematic view of a still further alternate configuration of a system for effectuating a methodology of fabricating lumen models in a microwell plate in accordance with the present invention;
(15) FIG. **13** is a cross-sectional view of the system of the present invention taken along line **13-13** of FIG. **12**;
(16) FIG. **14** is a top plan view of the system of FIG. **13**;
(17) FIG. **15** is a cross-sectional view of the system of the present invention, similar to FIG. **13**, showing first and second lumen models fabricated in the microwell plate; and
(18) FIG. **16** is a top plan view of the system of the present invention, similar to FIG. **14**, showing the first and second lumen models fabricated in the microwell plate.

DETAILED DESCRIPTION OF THE DRAWINGS

(19) Referring to FIGS. **1-5**, a system for effectuating the methodology of the present invention is generally designated by the reference numeral **5**. System **5** includes well plate **10** fabricated from a plastic material such as polyvinylchloride, polystyrene or polypropylene and has a generally rectangular configuration. However, other configurations are possible without deviating from the scope of the present invention. Well plate **10** is defined by first and second generally parallel ends **16** and **18**, respectively, interconnected first and second generally parallel sidewalls **20** and **22**, respectively, perpendicular thereto. Well plate **10** further includes an upwardly directed surface **24** and a downwardly directed surface **26**. A plurality of wells, designated collectively by the reference numeral **12**, are provided in upper surface **24** of well plate **10**. By way of example, in the depicted embodiment, well plate **10** includes sixty four (64) individual wells **14** arranged in eight (8) rows **30***a*-**30***h* and eight (8) columns **32***a*-**32***h*. However, it can be understood that the number and arrangement of wells **14** in well plate **10** may be varied, without deviating from the scope of the present invention.
(20) Each well **14** in upper surface **24** of well plate **10** is identical in structure. As such, the description hereinafter of well **14** located at the intersection of row **30***d* and column **32***b* is understood to described each of the plurality of wells **12**, as if fully described herein. Well **14** at the intersection of row **30***d* and column **32***b* is defined first and second generally parallel sidewalls **34** and **36**, respectively, generally parallel to first and second sides **20** and **22**, respectively, of well

plate **10** and generally perpendicular to first and second ends **16** and **18**, respectively, of well plate **10** and by third and fourth generally parallel sidewalls **38** and **40**, respectively, generally perpendicular to first and second sides **20** and **22**, respectively, of well plate **10** and generally parallel to first and second ends **16** and **18**, respectively, of well plate **10**. First face **42** of first sidewall **34**, first face **44** of second sidewall **36**, first face **46** of third sidewall **38** and first face **48** of fourth sidewall **40** are directed toward interior **50** of well **14**. Lower edges of first face **42** of first sidewall **34**, first face **44** of second sidewall **36**, first face **46** of third sidewall **38** and first face **48** of fourth sidewall **40** are interconnected by lower wall **56**.

(21) It can be understood that second face **51** of first sidewall **34** defines a corresponding first face **44** of second sidewall **36** in an adjacent well **14** in the same column, e.g. well **14** at row **30***c*, column **32***b*. Similarly, second face **53** of second sidewall **36** defines a corresponding first face **42** of first sidewall **34** in an adjacent well **14** in the same row, e.g. well **14** at row **30***e*, column **32***c*. Upper face **55** of first sidewall **34** extends between and interconnects upper edge **42***a* of first face **42** of first sidewall **34** and upper edge **51***a* of second face **51** of first sidewall **34**. Upper face **57** of second sidewall **44** extends between and interconnects upper edge **44***a* of first face **44** of second sidewall **36** and upper edge **53***a* of second face **53** of second sidewall **36**.

(22) Further, second face **52** of third sidewall **38** defines a corresponding first face in an adjacent well **14**, e.g. well **14** at row **30***a*, column **32***a*. Similarly, second face **54** of fourth sidewall **40** defines a corresponding first face in an adjacent well **14**, e.g. well **14** at row **30***a*, column **32***c*. Upper face **58** of third sidewall **38** extends between and interconnects upper edge **46***a* of first face **46** of third sidewall **38** and upper edge **52***a* of second face **52** of third sidewall **38**. Upper face **60** of fourth sidewall **40** extends between and interconnects upper edge **48***a* of first face **48** of fourth sidewall **40** and upper edge **54***a* of second face **54** of fourth sidewall **40**. It is noted that upper face **55** of first sidewall **34**, upper face **57** of second sidewall, upper face **58** of third sidewall **38** and upper face **60** of fourth sidewall **40** are co-planer and partially define upper surface **24** of well plate **10**.

(23) As best seen in FIG. **5**, first rod receiving slot **70** is provided in upper face **58** of third sidewall **38**. Rod receiving slot **70** is defined by first and second generally parallel sidewalls **72** and **74**, respectively, spaced from each other and extending between first and second faces **46** and **52**, respectively, thereof. First and second sidewalls **72** and **74**, respectively, terminate at and are interconnected by lower surface **76** spaced from and parallel to lower wall **56**. Lower surface **76** is generally perpendicular to first and second sidewalls **72** and **74**, respectively, and extends between first and second faces **46** and **52**, respectively, of third sidewall **38**.

(24) Similarly, referring to FIGS. **4-5**, second rod receiving slot **80** is provided in upper face **60** of fourth sidewall **40**. As hereinafter described, it is intended for second rod receiving slot **80** to axially aligned with and have an identical configuration to first rod receiving slot **70**. More specifically, second rod receiving slot **80** is defined by first and second generally parallel sidewalls **82** and **84**, respectively, spaced from each other and extending between first and second faces **48** and **54**, respectively, thereof. First and second sidewalls **82** and **84**, respectively, terminate at and are interconnected by lower surface **86** spaced from and parallel to lower wall **56**. Lower surface **86** is generally perpendicular to first and second sidewalls **82** and **84**, respectively, and extends between first and second faces **48** and **54**, respectively, of fourth sidewall **40**. It is intended for lower surfaces **76** and **86** partially defining corresponding first and second rod receiving slots **70** and **80**, respectively, to be co-planar and to define a lower limit as to the vertical spacing between lower wall **56** and the lumen model, hereinafter described.

(25) System **5** further includes fixture **100** to support and move a plurality of identical, parallelly extending rods **102**, as hereinafter described. It is for the plurality of rods **102** to be magnetic to facilitate the connection of the plurality of rods **102** to fixture **100**. However, the plurality of rods may be fabricated from other materials without deviating from the scope of the present invention. The plurality of rods **102** are spaced such that each rod **102** is axially aligned with a corresponding

row **30***a*-**30***h* of wells **14** in well plate **10**. Each of the plurality of rods **102** has an outer surface **104** and a generally circular cross-section. It is intended for fixture **100** to simultaneously move the plurality of rods **102** axially from a first retracted position, shown in phantom in FIG. **1**, wherein the plurality of rods **102** are removed from well plate **10**, and second extended position wherein each rod **102** extends through first and second rod receiving slots **70** and **80**, respectively, in third and fourth sidewalls **38** and **40**, respectively, defining each well **14** in a corresponding row **30***a*-**30***h* of wells **14** in well plate **10**, FIGS. **1-3**.

(26) In operation, with the plurality of rods **102** in the first retracted position, fixture **100** and well plate **10** are aligned such that each rod **102** of the plurality of rods **102** in axially aligned with first and second rod receiving slots **70** and **80**, respectively, in third and fourth sidewalls **38** and **40**, respectively, defining each well **14** in a corresponding row **30***a*-**30***h* of wells **14** in well plate **10**. Thereafter, the plurality of rods **102** are moved from the first retracted position to the second extended position wherein each rod **102** extends through first and second rod receiving slots **70** in third and fourth sidewalls **38** and **40**, respectively, defining each well **14** in a corresponding row **30***a*-**30***h* of wells **14** in well plate **10**.

(27) Referring back to FIGS. **4-5**, unpolymerized, polymerizable material **108**, e.g. a synthetic hydrogel, is deposited into interiors **50** of all of or a portion of wells **14** in well plate **10**, e.g., wells **14** in columns **32***b* and **32***f* of well plate **10** in any conventional manner, such as by pipetting. With polymerizable material **108** deposited into interiors **50** of wells **14** in columns **32***b* and **32***f* of well plate **10**, a predetermined stimulus, e.g., heat or light, is directed at material **108** in interiors **50** of wells **14** in columns **32***b* and **32***f* of well plate **10** such that material **108** in the interiors **50** of wells **14** in columns **32***b* and **32***f* is polymerized. With material **108** in the interior **50** of wells **14** in columns **32***b* and **32***f* polymerized, fixture **100** retracts the plurality of rods **102** from the polymerized material **108** in interior **50** of wells **14** in columns **32***b* and **32***f*, such that a lumen model or generally tubular passageway **112** extends through polymerized material **108** extending between third and fourth sidewalls **38** and **40**, respectively. Tubular passageway **112** is defined by tubular surface **114** of polymerized material **108** and includes a first opening **116** communicating with first rod receiving slot **70** in third sidewall **38** and a second opening communicating with second rod receiving slot **80** in fourth sidewall **40**.

(28) It can be understood that a first media, which may include cells, molecules or the like, may be deposited in a well, e.g., well **14** at row **30***d*, column **32***a*, adjacent to a corresponding well, e.g., well **14** at row **30***d*, column **32***b*, having tubular passageway **112** extending through polymerized material **108** therein. The first media in well **14** at row **30***d*, column **32***a* communicates with first opening **116** through first rod receiving slot **70** in third sidewall **38** of well **14** at row **30***d*, column **32***b*. As such, the first media travels through first opening **116**, tubular passageway **112**, second opening **118** and second rod receiving slot **80** in fourth sidewall **40** into well **14**, e.g., at row **30***d*, column **32***c*, adjacent thereto by means of gravity, a mechanical device or the like.

(29) Similarly, a second media, which may include cells, molecules or the like, may be deposited in a well, e.g., well **14** at row **30***d*, column **32***e*, adjacent to a corresponding second well, e.g., well **14** at row **30***d*, column **32***f*, having tubular passageway **112** extending through polymerized material **108** therein. The second media in well **14** at row **30***d*, column **32***e* communicates with first opening **116** through first rod receiving slot **70** in third sidewall **38** well **14** at row **30***d*, column **32***f*. As such, the second media travels through first opening **116**, tubular passageway **112**, second opening **118** and second rod receiving slot **80** in fourth sidewall **40** into well **14**, e.g., at row **30***d*, column **32***g*, adjacent thereto by means of gravity, a mechanical device or the like.

(30) In can be appreciated that utilizing the same methodology heretofore described, the same or different media may be flowed through each tubular passageway **112** extending through the portion of the plurality of wells **14** having polymerized material **108** therein, thereby allowing a user to run multiple tests and/or experiments simultaneously. It is noted that wells **14**, e.g., wells **14** in column **30***d*, may be used to isolate and separate the portion of the plurality of wells **14** having polymerized

material **108** therein from each other and from the media introduced therein.
(31) Referring to FIGS. **6-11**, it is contemplated to provide additional lumen models or generally tubular passageways through polymerized material **108**. More specifically, third rod receiving slot **120** is provided in upper face **55** of first sidewall **34**. Third rod receiving slot **120** is defined by first and second generally parallel sidewalls **122** and **124**, respectively, spaced from each other and extending between first and second faces **42** and **51**, respectively, first sidewall **34**. First and second sidewalls **122** and **124**, respectively, terminate at and are interconnected by lower surface **126** spaced from and parallel to lower wall **56**. Lower surface **126** is generally perpendicular to first and second sidewalls **122** and **124**, respectively, and extends between first and second faces **42** and **51**, respectively, of first sidewall **34**.
(32) Similarly, fourth rod receiving slot **130** is provided in upper face **57** of second sidewall **36**. For reasons hereinafter described, it is intended for fourth rod receiving slot **130** to axially aligned with and have an identical configuration to third rod receiving slot **120**. More specifically, fourth rod receiving slot **130** is defined by first and second generally parallel sidewalls **132** and **134**, respectively, spaced from each other and extending between first and second faces **44** and **53**, respectively, of second sidewall **36**. First and second sidewalls **132** and **134**, respectively, terminate at and are interconnected by lower surface **136** spaced from and parallel to lower wall **56**. Lower surface **136** is generally perpendicular to first and second sidewalls **132** and **134**, respectively, and extends between first and second faces **44** and **53**, respectively, of second sidewall **36**. It is intended for lower surfaces **126** and **136** defining partially defining third and fourth rod receiving slots **120** and **130**, respectively, to be co-planar and to define a lower limit as to the vertical spacing between lower wall **56** and the lumen model, as hereinafter described.
(33) System **5** may further include second fixture **140** supporting a second plurality of identical, parallelly extending rods **142**, as hereinafter described, FIG. **6**. It is for the plurality of rods **142** to be magnetic to facilitate the connection of the plurality of rods **142** to second fixture **140**. However, the plurality of rods **142** may be fabricated from other materials without deviating from the scope of the present invention. The plurality of rods **142** are spaced such that each rod **142** is axially aligned with a corresponding columns **32***a*-**32***h* of wells **14** in well plate **10**. Each of the plurality of rods **142** has an outer surface **144** and a generally circular cross-section. It is intended for fixture **140** to simultaneously move the plurality of rods **142** axially from a first retracted position, shown in phantom in FIG. **6**, wherein the plurality of rods **142** are removed from well plate **10**, and second extended position wherein each rod **142** extends through third and fourth rod receiving slots **120** and **130**, respectively, in first and second sidewalls **34** and **36**, respectively, defining each well **14** in a corresponding column **32***a*-**32***h* of wells **14** in well plate **10**.
(34) In operation, with the plurality of rods **102** and the plurality of rods **142** in the first retracted positions, fixture **100** and well plate **10** are aligned such that each rod **102** of the plurality of rods **102** in axially aligned with first and second rod receiving slots **70** and **80**, respectively, in third and fourth sidewalls **38** and **40**, respectively, defining each well **14** in a corresponding row **30***a*-**30***h* of wells **14** in well plate **10**. Similarly, second fixture **140** and well plate **10** are aligned such that each rod **142** of the plurality of rods **142** is axially aligned with third and fourth second rod receiving slots **120** and **130**, respectively, in first and second sidewalls **34** and **36**, respectively, defining each well **14** in a corresponding columns **32***a*-**32***h* of wells **14** in well plate **10**. It is noted that the plurality of rods **102** lie in a first plane parallel to and vertically spaced from a plane in which lower walls **56** of the plurality of wells **14** lie and the plurality of rods **142** lie in a second plane parallel to and vertically spaced from the first plane and from the plane in which lower walls **56** of the plurality of wells **14** lie, thereby allowing the plurality of rods **102** and the plurality of rods **142** to cross each other within the plurality of wells **14** of well plate **10** when the plurality of rods **102** and the plurality of rods **142** are moved to the second extended positions, FIGS. **7-8**.
(35) As heretofore described, the plurality of rods **102** are moved from the first retracted position to the second extended position wherein each rod **102** extends through first and second rod receiving

slots **70** in third and fourth sidewalls **38** and **40**, respectively, defining each well **14** in a corresponding row **30***a*-**30***h* of wells **14** in well plate **10**. Similarly, the plurality of rods **142** are moved from the first retracted position to the second extended position wherein each rod **142** extends through third and fourth second rod receiving slots **120** and **130**, respectively, in first and second sidewalls **34** and **36**, respectively, defining each well **14** in a corresponding columns **32***a*-**32***h* of wells **14** in well plate **10**. In the depicted embodiment, the lower portions of each of the plurality of rods **142** engage corresponding upper portions of the plurality of rods **102** in each row **30***a*-**30***h* within interiors **50** of wells **14** in each row **30***a*-**30***h* of wells **14** in well plate **10**.

(36) Unpolymerized, polymerizable material **108**, such as a synthetic hydrogel, is deposited into interior **50** of all or a portion of the plurality of wells **14** in well plate **10**, e.g., via pipetting. For example, it is contemplated to provide unpolymerized, polymerizable material **108** in the interior **50** of well **14** at the intersection of row **30***d* with column **32***b*. Thereafter, a predetermined stimulus, e.g., heat or light, is directed at material **108** in the interior **50** of well **14** at the intersection of row **30***d* with column **32***b* such that material **108** in the interior **50** of well **14** at the intersection of row **30***d* with column **32***b* is polymerized. With material **108** in the interior **50** of well **14** at the intersection of row **30***d* with columns **32***b* polymerized, fixture **100** retracts the plurality of rods **102** from the polymerized material **108** in interior **50** of well **14** at the intersection of row **30***d* with column **32***b* such that a lumen model or generally tubular passageway **112**, as heretofore described, extends through polymerized material **108** between third and fourth sidewalls **38** and **40**, respectively. Similarly, second fixture **140** retracts the plurality of rods **142** from the polymerized material **108** in interior **50** of well **14** at the intersection of row **30***d* with column **32***b* such that a lumen model or generally tubular passageway **152** extends through polymerized material **108** between first and second sidewalls **34** and **36**, respectively. Tubular passageway **152** is defined by tubular surface **154** of polymerized material **108** and includes a first opening **156** communicating with third rod receiving slot **120** in first sidewall **34** and a second opening **158** communicating with fourth rod receiving slot **130** in second sidewall **36**. In the depicted embodiment, tubular passageways **112** and **152** communicate with each other at intersection **160**, FIG. **11**, and extend along axes generally perpendicular to each other. It can be appreciated the axes along which tubular passageways **112** and **152** extend may be vertically adjusted, and spaced from each other, by adjusting the vertical positions of the plurality of rods **102** and the plurality of rods **142**, respectively.

(37) It can be understood that a first media, which may include cells, molecules or the like, may be deposited in a well, e.g., well **14** at row **30***d*, column **32***a*, adjacent to a corresponding well, e.g., well **14** at row **30***d*, column **32***b*, having tubular passageway **112** extending through polymerized material **108** therein. As such, the first media travels through first opening **116**, tubular passageway **112**, second opening **118** and second rod receiving slot **80** in fourth sidewall **40** into well **14**, e.g., at row **30***d*, column **32***c*, adjacent thereto by means of gravity, a mechanical device or the like.

(38) Similarly, a second media, which may include cells, molecules or the like, may be deposited in a well, e.g., well **14** at row **30***c*, column **32***b*, adjacent to a corresponding well **14** at row **30***d*, column **32***b*, having tubular passageway **112** extending through polymerized material **108** therein. As such, the second media travels through first opening **156**, tubular passageway **152**, second opening **158** and fourth rod receiving slot **130** in second sidewall **36** into well **14** at row **30***e*, column **32***b*, adjacent thereto by means of gravity, a mechanical device or the like. It can appreciated that the first and second media may communicate with each other through intersection **160** of tubular passageways **112** and **152**.

(39) In can be appreciated that utilizing the same methodology heretofore described, the same or different media may be flowed through each tubular passageways **112** and **152** extending through the portion of the plurality of wells **14** having polymerized material **108** therein, thereby allowing a user to run multiple tests and/or experiments simultaneously.

(40) In addition, referring to FIGS. **12-16**, it is contemplated to provide additional lumen models or

generally tubular passageways parallel to tubular passageway **112** in polymerized material **108**. Fifth rod receiving slot **170** is provided in upper face **58** of third sidewall **38**. Fifth rod receiving slot **170** is defined by first and second generally parallel sidewalls **172** and **174**, respectively, spaced from each other and extending between first and second faces **46** and **52**, respectively, thereof. First and second sidewalls **172** and **174**, respectively, terminate at and are interconnected by lower surface **176** spaced from and parallel to lower wall **56**. Lower surface **176** is generally perpendicular to first and second sidewalls **172** and **174**, respectively, and extends between first and second faces **46** and **52**, respectively, of third sidewall **38**.

(41) Similarly, sixth rod receiving slot **180** is provided in upper face **60** of fourth sidewall **40**. As hereinafter described, it is intended for second rod receiving slot **180** to axially aligned with and have an identical configuration to first rod receiving slot **170**. More specifically, sixth rod receiving slot **180** is defined by first and second generally parallel sidewalls **182** and **184**, respectively, spaced from each other and extending between first and second faces **48** and **54**, respectively, thereof. First and second sidewalls **182** and **184**, respectively, terminate at and are interconnected by lower surface **186** spaced from and parallel to lower wall **56**. Lower surface **186** is generally perpendicular to first and second sidewalls **182** and **184**, respectively, and extends between first and second faces **48** and **54**, respectively, of fourth sidewall **40**. It is intended for lower surfaces **176** and **186** partially defining corresponding first and second rod receiving slots **170** and **180**, respectively, to be co-planar and to define a lower limit as to the vertical spacing between lower wall **56** and the lumen model, hereinafter described.

(42) In addition to supporting and moving the plurality of rods **102**, as heretofore described, it is contemplated for fixture **100** to support and move a second plurality of identical, parallelly extending rods **192**, as hereinafter described. It is for the second plurality of rods **192** to be magnetic to facilitate the connection of the plurality of rods **192** to fixture **100**. However, the plurality of rods **192** may be fabricated from other materials without deviating from the scope of the present invention. It is contemplated for the plurality of rods **192** to lie in a common horizontal plane and be spaced from plurality of rods **102** or, as depicted in FIG. **13**, to be vertically spaced from the plurality of rods **102** and to be spaced such from each other such that each rod **192** is axially aligned with a corresponding row **30***a*-**30***h* of wells **14** in well plate **10**. Each of the plurality of rods **192** has an outer surface **194** and a generally circular cross-section. It is intended for fixture **100** to simultaneously move the plurality of rods **102** and the second plurality of rods **192** axially from a first retracted position, shown in phantom in FIG. **1**, wherein the plurality of rods **102** and the second plurality of rods **192** are removed from well plate **10**, and second extended position wherein: 1) each rod **102** extends through first and second rod receiving slots **70** and **80**, respectively, in third and fourth sidewalls **38** and **40**, respectively, defining each well **14** in a corresponding row **30***a*-**30***h* of wells **14** in well plate **10**; and 2) each rod **192** extends through first and second rod receiving slots **170** and **180**, respectively, in third and fourth sidewalls **38** and **40**, respectively, defining each well **14** in a corresponding row **30***a*-**30***h* of wells **14** in well plate **10**In operation, with the plurality of rods **102** and the second plurality of wells **192** in the first retracted position, fixture **100** and well plate **10** are aligned such that each rod **102** of the plurality of rods **102** in axially aligned with first and second rod receiving slots **70** and **80**, respectively, in third and fourth sidewalls **38** and **40**, respectively, defining each well **14** in a corresponding row **30***a*-**30***h* of wells **14** in well plate **10** and such that each rod **192** of the second plurality of rods **192** in axially aligned with first and second rod receiving slots **170** and **180**, respectively, in third and fourth sidewalls **38** and **40**, respectively, defining each well **14** in a corresponding row **30***a*-**30***h* of wells **14** in well plate **10**. Thereafter, the plurality of rods **102** and the second plurality of rods **192** are moved from the first retracted position to the second extended position wherein: 1) each rod **102** extends through first and second rod receiving slots **70** and **80**, respectively, in third and fourth sidewalls **38** and **40**, respectively, defining each well **14** in a corresponding row **30***a*-**30***h* of wells **14** in well plate **10**; and 2) each rod **192** extends through first and second rod receiving slots **170**

and **180**, respectively, in third and fourth sidewalls **38** and **40**, respectively, defining each well **14** in a corresponding row **30***a*-**30***h* of wells **14** in well plate **10**.
(43) Unpolymerized, polymerizable material **108**, such as a synthetic hydrogel, is deposited into interior **50** of a portion of wells **14** in each row **30***a*-**30***h* of wells **14** in well plate **10**, e.g., via pipetting. For example, it is contemplated to provide unpolymerized, polymerizable material **108** in the interior **50** of well **14** at the intersection of row **30***d*, column **32***b*. Thereafter, a predetermined stimulus, e.g., heat or light, is directed at material **108** in the interior **50** of well **14** at the intersection of row **30***d*, column **32***b* such that material **108** in the interior **50** of well **14** at the intersection of row **30***d*, column **32***b* is polymerized. With material **108** in the interior **50** of well **14** at the intersection of row **30***d*, column **32***b* polymerized, fixture **100** retracts the plurality of rods **102** and the second plurality of rods **192** from the polymerized material **108** in interior **50** of well **14** at the intersection of row **30***d*, column **32***b*, such that in addition to generally tubular passageway **112** extending through polymerized material **108** extending between third and fourth sidewalls **38** and **40**, respectively, a second, generally tubular passageway **202** extends through polymerized material **108** extending between third and fourth sidewalls **38** and **40**, respectively. Second tubular passageway **202** is defined by tubular surface **204** of polymerized material **108** and includes a first opening **206** communicating with fifth rod receiving slot **170** in third sidewall **38** and a second opening **208** communicating with sixth rod receiving slot **180** in fourth sidewall **40**.
(44) It can be understood that a media, which may include cells, molecules or the like, may be deposited in well **14** at row **30***d*, column **32***a*, adjacent to well **14** at row **30***d*, column **32***b*, having tubular passageways **112** and **202** extending through polymerized material **108** therein. As such, the media travels through: 1) first opening **116**, tubular passageway **112**, second opening **118** and second rod receiving slot **80** in fourth sidewall **40** into well **14**, e.g., at row **30***d*, column **32***c*, adjacent thereto by means of gravity, a mechanical device or the like; and 2) first opening **206**, second tubular passageway **202**, second opening **208** and fourth rod receiving slot **180** in fourth sidewall **40** into well **14** at row **30***d*, column **32***c*, adjacent thereto by means of gravity, a mechanical device or the like.
(45) As described, it can be understood that the description of system **5** is merely exemplary and that numerous variations are possible without deviating from the scope of the present invention. For example, it can appreciated that the configuration, arrangement and number of wells **14** in well plate **10** may be varied without deviating from the scope of the present invention. Further, it can be understood that the arrangement, number and orientation of the tubular passageways through the polymerized material in the wells **14** in well plate **10** may be varied without deviating from the scope of the present invention. Likewise, it can be appreciated that the arrangement, number and orientation of the tubular passageways through the polymerized material in each wells **14** in well plate **10** may be the same or different, by altering the arrangement, number and orientation of the plurality of rods passing through each well **14** in well plate **10**.
(46) Various modes of carrying out the invention are contemplated as being within the scope of the following claims particularly pointing out and distinctly claiming the subject matter that is regarded as the invention.

Claims

1. A device for creating a lumen in a microenvironment, comprising: a microwell plate defining a plurality of wells arranged in rows and columns, each of the plurality of wells defined by at least one sidewall having first and second openings therein; a plurality of rods, each rod selectively moveable between a first position wherein each rod is removed from corresponding first and second openings in each of the at least one sidewall defining the plurality of wells and a second position wherein each rod is inserted through the corresponding first and second openings in each of the at least one sidewall defining the plurality of wells; and a fixture interconnected to the

plurality of rods, the fixture configured for moving the plurality of rods between the first and second positions.
2. The device of claim 1 wherein each of the plurality of rods are parallel to each other.
3. The device of claim 1 wherein: the at least one sidewall defining the plurality of wells has third and fourth openings therein; the plurality of rods is a first plurality of rods; and the device further includes a second plurality of rods, each rod of the second plurality of rods selectively moveable between a first position wherein each rod of the second plurality of rods is removed from the third and fourth openings in each of the at least one sidewall defining the plurality of wells and a second position wherein each rod of the second plurality of rods is inserted through the third and fourth openings in each of the at least one sidewall defining the plurality of wells.
4. The device of claim 3 wherein the fixture is interconnected to the second plurality of rods, the fixture configured for moving the second plurality of rods between the first and second positions.
5. The device of claim 3 wherein the first plurality of rods lie in a first plane and the second plurality of rods lie in a second plane spaced from the first plane.
6. The device of claim 3 wherein the fixture is a first fixture and wherein the device further comprises a second fixture interconnected to the second plurality of rods, the second fixture configured for moving the second plurality of rods between the first and second positions.
7. The device of claim 3 wherein the first plurality of rods is transverse to the second plurality of rods.
8. The device of claim 3 wherein: each of the first plurality of rods travels along a path between the first and second positions of the first plurality of rods; each of the second plurality of rods travels along a path between the first and second positions of the second plurality of rods; and the path of each of the first plurality of rods is interested by the paths of the second plurality of rods.
9. The device of claim 1 wherein each of the plurality of rods has a circular cross-section.
10. The device of claim 2 wherein each of the plurality of rods is magnetic.
11. The device of claim 3 wherein each of the plurality of rods is fabricated from a polymer.